

Title: Beyond Transformers

Subtitle: Architectures for the Next Generation ML and GenAI Systems

V_1_1

ABOUT THE AUTHOR

Raphaël MANSUY

Raphaël Mansuy is a French AI entrepreneur and strategist based in Hong Kong, specializing in Agentic AI, data engineering, and systems that bridge software engineering with autonomous intelligence. With over two decades of global technology leadership, he has been recognized among the world's top 100 most influential people in AI.

As CTO of ELITIZON, a global AI consulting firm based in Hong Kong, Mansuy leads teams across Europe and Asia in deep learning, NLP, MLOps, and computer vision. He founded Quantalogic, a sovereign AI Agent platform for Europe, and StudentCentral.ai, an initiative reimagining education through AI. A proponent of building in public, he advises major corporations on data strategy and AI transformation from his unique East-West vantage point.

Annaëlle MANSUY

Annaëlle Mansuy is a computer science student pursuing a Master of Engineering at University College London (UCL), with a focus on machine learning, generative AI, and software development. She has hands-on experience building applications powered by large language models and vision AI.

As an AI Engineer at ELITIZON, she has contributed to projects including structured content extraction, audio-to-text transcription systems featuring diarization and retrieval-augmented generation (RAG) for fact-checking. Her work spans the full software development lifecycle—from requirements and testing to backend, frontend, and integration.

With a foundation in mathematical finance, including financial optimization and econometrics, Annaëlle brings an interdisciplinary perspective to her technical work. She is currently preparing to pursue research in machine learning.

The Book's Goal

What kind of individual would be interested in this book?	<i>Senior Machine Learning Engineers, AI Researchers, Systems Architects, and CTOs who are hitting the efficiency and reasoning limits of standard Transformers.</i>
What knowledge do they need before they start reading?	<i>A solid understanding of Deep Learning fundamentals (PyTorch/JAX), familiarity with the Transformer architecture (Attention, KV-Cache), and experience deploying or fine-tuning LLMs.</i>
Why should they buy this book?	<i>To future-proof their career and their tech stack. This is the first book to move beyond "Prompt Engineering" and teach the engineering physics of the next wave of AI (Mamba, RWKV, JEPA, and Neuro-symbolic systems).</i>
What is the product approach and USP of the book?	<i>USP: It is an "Architectural Intuition Builder." Instead of just surveying papers, it defines a taxonomy (Generative vs. Predictive vs. Energy-Based) and uses 4 consistent "Running Examples" to prove which architecture solves which problem.</i>
Product Breakdown: In two sentences, describe the "journey" the book takes the reader on. Look at your section headings for help	<i>The reader starts by mathematically dissecting why Transformers fail at scale (the "Wall"), then systematically builds competence in every major alternative (Recurrent, State-Space, Hybrid, Energy-Based, and JEPA). Finally, they learn to combine these into unified, agentic systems that are efficient enough to run on the edge and smart enough to reason/plan.</i>
By the end of this book, you will...	<i>Be able to select, architect, and optimize the correct post-Transformer model (SSM, Hybrid, or World Model) for any</i>

	<i>specific constraint—be it infinite context, millisecond latency, or verifiable reasoning.</i>
--	--

COMPETITIVE BOOK TITLES

List the books here:

No.	Title	Opportunities for Packt (gaps, poor reviews, less-popular authors)
1	<i>"Generative AI with LangChain"</i> (Packt/Competitor)	Gap: Focuses entirely on tooling/orchestration of <i>existing</i> APIs (OpenAI), not the underlying <i>model architecture</i> . Our book targets the builders of the <i>next</i> models, not just the users of current ones.
2	<i>"Build a Large Language Model (From Scratch)"</i> (Manning)	Gap: Excellent for understanding the <i>baseline</i> (Transformers), but stops there. It does not cover Mamba, JEPA, or Neuro-symbolic hybrids, leaving readers unprepared for 2026.
3	<i>"Designing Machine Learning Systems"</i> (O'Reilly)	Gap: High-level systems focus. It lacks the mathematical depth on <i>specific</i> new architectures (like Selective Scan or WKV kernels) that engineers need to implement these systems.

LEARNING OUTCOME - WHAT WILL THE READER LEARN AND DO?

Decide what the key learning objectives will be for your book. Please, list them below.

Also, consider the competing books; in particular the **description**, **table of contents** and **book reviews**, how do the learning objectives of this book make it unique?

1	Diagnose Transformer Limits: Mathematically calculate the memory bandwidth and FLOPs bottlenecks of Attention for any given sequence length to determine when to switch architectures.
2	Master Linear Architectures: Implement and train Gated Linear Recurrent models (xLSTM, RWKV) and State Space Models (Mamba) from scratch using PyTorch.
3	Design Hybrid Systems: Architect "Best of Both Worlds" models by interleaving SSM layers (for bulk context) with Attention layers (for recall) to optimize the accuracy-latency curve.
4	Implement Neuro-Symbolic Reasoning: Build "System 2" agents that use Graph Neural Networks (GNNs) and Energy-Based Models (EBMs) to verify facts and constrain hallucinations.
5	Deploy World Models: Construct Joint Embedding Predictive Architectures (JEPA) that plan in latent space, achieving efficiency gains for robotic or video tasks.
6	Optimize for Hardware: Write custom CUDA/Triton kernels for Selective Scan and WKV operations to maximize inference throughput on consumer GPUs.
7	Evaluate Beyond Perplexity: Apply a new suite of benchmarks to measure "Long-Context Degradation" and "Agentic Planning Success" rather than just next-token prediction.
8	

Parts and Chapters

Chapter Outline

Part 1: The Baseline and the Bottleneck	
1.	The Transformer Paradigm (2017–2023)
2.	The Architectural Taxonomy of 2026
3.	

Part 2: Linear-Time & Recurrent Architectures	
1.	Modern Recurrent Neural Networks (The Gated Linear Pattern)
2.	State Space Models (SSMs)
3.	Comparative Dynamics

Part 3: Compositional & Hybrid Architectures	
1.	The Hybrid Pattern (Attention + State)

2.	Multimodal Hybrids (Vision & Video)
3.	

Part 4: Neuro-Symbolic, World Models, and Reasoning

1.	Graph Neural Networks (GNNs) & Structured Reasoning
2.	Energy-Based Models (EBMs) & Dynamic Inference
3	World Models & Joint Embedding Architectures (JEPA)
4	Agentic Loops & Hierarchical Planning

Part 5: Efficiency, Training, & Production

1.	Extreme Compression & Quantization
2.	Hardware-Aware Design
3	Evaluation & Benchmarks (2025-2026)

Part 6: Future Directions

1.	Toward Unified Architectures (2026–2030)
2.	Societal & Ethical Implications

Detailed Outline

Now it's time to plan out your chapters in more detail. In the following section you will have the opportunity to list what each chapter does and what the user is gaining from it. The first one has been filled out as an example. Use this as a template and add as many additional chapters as you need.

Chapter 1: The Transformer Paradigm (2017–2023)

40 pages

Description: This chapter establishes the engineering baseline by rigorously analyzing the Transformer architecture. It moves beyond a basic introduction to focus on the "Physics of Attention"—specifically why the $O(N^2)$ complexity and KV-cache growth create a hard wall for long-context applications. It introduces the first Running Example, The Efficient Scalar.

Chapter Subheadings

- **The Attention Mechanism:** Inductive bias and global correlation.
- **The Hardware Lottery:** Why GPUs favored Transformers over RNNs.
- **The Inference Asymmetry:** Analyzing the "Prefill vs. Decode" compute discrepancy.
- **The Wall:** Case Study—Calculating memory bandwidth for 128k context on *The Efficient Scalar*.

Chapter 2: The Architectural Taxonomy of 2026

30 pages

Description: A high-level guide to the post-Transformer landscape. This chapter teaches readers how to classify novel architectures into three buckets: Generative (Explicit), Predictive (Latent), or Energy-Based (Implicit). It provides a decision framework for choosing the right tool.

Chapter Subheadings

- **Mapping the Landscape:** Recurrent vs. State-Space vs. Energy-Based.
- **Benchmarking the Shift:** 2025 Comparative analysis (Throughput vs. Perplexity).
- **Design Intuition:** When to choose State (Compression) vs. Attention (Retrieval).
- **The 4 Canonical Examples:** Defining the running use-cases (Scalar, Agent, Reasoner, Streamer).

Chapter 3: Modern Recurrent Neural Networks

45 pages

Description: This chapter explores the resurgence of RNNs through the "Gated Linear Recurrence" pattern. It deep dives into xLSTM and RWKV-7, showing how they solve the vanishing gradient problem and achieve parallelized training via WKV kernels.

Chapter Subheadings

- **xLSTM:** sLSTM/mLSTM blocks and exponential gating.
- **RWKV-7 "Goose":** Expressive dynamic state evolution.
- **The WKV Kernel:** Parallelizing RNN training on GPUs
- **Implementation:** Building *The Efficient Scalar* with RWKV to reduce VRAM by 40%.

Chapter 4: State Space Models (SSMs)

50 pages

Description: A comprehensive guide to Mamba and its descendants. It explains the transition from Continuous Time Control (S4) to Discrete Deep Learning, focusing heavily on the "Selectivity Mechanism" that allows Mamba to filter noise and scale linearly.

Chapter Subheadings

- **From S4 to Mamba:** The discretization process explained.
- **The Selectivity Breakthrough:** How to learn *what to forget*.
- **Scaling SSMs:** Routing Mamba (MoE) and Mixture-of-Mamba.
- **Case Study:** Benchmarking *The Long-Context Agent* on "Needle in a Haystack".

Chapter 5: Comparative Dynamics

30 pages

Description: This chapter unifies the math, showing how SSMs, Linear Attention, and RNNs are cousins. It crucially analyzes the "Trade-offs"—specifically proving why state-based models struggle with "Associative Recall" compared to Global Attention.

Chapter Subheadings

- **The Unification:** Mathematical equivalence of Linear Attention and SSMs.
- **The Copying Problem:** Why State models struggle with exact recall.
- **Trade-off Analysis:** Latency vs. Accuracy curves.
- **RetNet:** The Exponential Moving Average (EMA) family.

Chapter 6: The Hybrid Pattern (Attention + State)

40 pages

Description: This chapter teaches the practical engineering of Hybrid models (like Samba and Griffin). Readers learn how to interleave SSM layers (for bulk processing) with sliding-window Attention (for precision) to build systems that handle infinite context without losing detail.

Chapter Subheadings

- **Samba & Griffin:** The "Sliding Window" architecture.
- **Architectural Placement:** Early, Middle, or Late Attention?
- **Memory Management:** Balancing KV-cache with Recurrent State.
- **Implementation:** Hybridizing *The Long-Context Agent*.

Chapter 7: Multimodal Hybrids (Vision & Video)

35 pages

Description: Moving beyond text, this chapter explores how linear architectures are revolutionizing Vision. It covers MambaVision and VAMBA, showing how 1D sequence models can process 2D/3D data more efficiently than ViTs or 3D-UNets.

Chapter Subheadings

- **MambaVision:** Replacing ViT with Hybrid Stacks.
- **VAMBA:** Temporal coherence in video generation.
- **The 2D Scan:** Serialization strategies for images.
- **Case Study:** *The Dynamic Streamer* at 60fps.

Chapter 8: Graph Neural Networks (GNNs) & Structured Reasoning

40 pages

Description: This chapter addresses the "Structure Gap." It explains why sequence models fail at non-linear relationships and how to use GNNs as a "Relational Inductive Bias." It introduces the *ConstraintLLM* framework for Neuro-Symbolic integration.

Chapter Subheadings

- **The Structure Gap:** Why LLMs fail at topology.
- **Graph-LLM Hybrids:** Encoding Knowledge Graphs (KGs) into Latent Space.
- **ConstraintLLM:** Enforcing symbolic logic on neural outputs.
- **Implementation:** Grounding *The Structural Reasoner* in an Enterprise KG.

Chapter 9: Energy-Based Models (EBMs) & Dynamic Inference

45 pages

Description: A shift from "System 1" (Reflexive) to "System 2" (Reflective). This chapter introduces Energy-Based Models, where generation is an optimization process. It covers Energy-Based Transformers (EBTs) that "ponder" via internal gradient descent before generating.

Chapter Subheadings

- **Implicit Generation:** $P(x)$ vs. Energy Functions.
- **Energy-Based Transformers (EBTs):** The "Think Loop" at inference.
- **Neuro-Symbolic EBMs:** Logic as Energy Constraints.
- **Case Study:** Self-correcting math errors in *The Efficient Scalar*.

Chapter 10: World Models & Joint Embedding Architectures (JEPA)

50 pages

Description: The "Predictive" paradigm. This chapter explains why "Prediction is not Generation." It details LeCun's JEPA architecture, which predicts latent state evolution rather than pixels, enabling efficient planning and reasoning.

Chapter Subheadings

- **Prediction vs. Generation:** The Pixel-Level fallacy.
- **LeJEPA & LLM-JEPA:** Learning abstract representations.
- **The Planning Engine:** Using V-JEPA for robotic simulation.
- **Implementation:** *The Dynamic Streamer* with a Latent Predictor.

Chapter 11: Agentic Loops & Hierarchical Planning

50 pages

Description: Bringing it all together into autonomous agents. This chapter focuses on "Hierarchical Decision Mamba" and other architectures designed for long-horizon loops, contrasting them with the brittleness of Transformer-based agents.

Chapter Subheadings

- **Hierarchical State:** Managing long time horizons.
- **Nemotron Nano:** Efficient on-device agency.
- **The Agent Stack:** JEPA (Planner) + EBM (Verifier) + Mamba (Actor).
- **Case Study:** An autonomous coding loop for *The Long-Context Agent*.

Chapter 12: Extreme Compression & Quantization

30 pages

Description: Addressing the unique challenges of compressing state-based models. This chapter covers Bi-Mamba (1-bit architectures) and the sensitivity of recurrent weights to quantization noise.

Chapter Subheadings

- **State Sensitivity:** Why RNNs are harder to quantize than Transformers.
- **Bi-Mamba: 1-Bit State Space Models.**
- **Sparse Routing:** Pruning connections in MoE Mambas.
- **Implementation:** *The Dynamic Streamer* with a Latent Predictor.
- **Practical Guide:** Quantizing RWKV for mobile deployment.

Chapter 13: Evaluation & Benchmarks (2025-2026)

30 pages

Description: How to measure success when "Perplexity" is no longer enough. This chapter introduces new metrics for Long-Context recall, Agentic success rates, and recursive extrapolation.

Chapter Subheadings

- **Beyond Perplexity:** The reasoning gap.
- **The Needle & The Haystack:** Measuring context degradation.
- **Agentic Benchmarks:** Evaluating planning vs. luck.
- **Beyond GPUs:** Neuromorphic hardware trends.

- **Reference:** The 2026 Leaderboard.

Chapter 14: Toward Unified Architectures (2026–2030)

25 pages

Description: A forward-looking chapter on the convergence of these architectures. It speculates on "The Grand Unification" of SSMs and Attention, and the role of Recursive "Tiny Networks" in self-improving AI.

Chapter Subheadings

- **Grand Unification:** Mathematical convergence.
- **Recursive Reasoning:** "Less is More" with Tiny Networks.
- **Self-Improvement:** Bootstrapping intelligence.
- **The Road to 2030:** Predictions.

Chapter 15: Societal & Ethical Implications

20 pages

Description: A sober analysis of the risks and benefits. It focuses on the "Democratization via Efficiency" (running powerful models on cheap hardware) and the specific safety risks of autonomous agentic loops.

Chapter Subheadings

- **Democratization:** AI on the Edge.
- **Agentic Risks:** When loops go wrong.
- **Neuro-Symbolic Safety:** Verifiability and Trust.
- **Conclusion:** Building a responsible future.

– end of outline –

Community outreach (optional)

Technical Reviewers

Can you recommend peers and members of your community to become technical reviewers?

Technical Reviewers (*Ideally, select experts from the RWKV, Mamba, or LeCun Lab communities*)

Full name	Email Address	LinkedIn Profile
Tri Dao (Mamba Author)		
Bo Peng (RWKV Author)		
Yann LeCun (JEPA Advocate)		

Amazon Reviewers

Full name	Email Address	LinkedIn Profile

Influencers

Can you recommend any influential community members or organizations for Packt to collaborate with on the marketing campaign of your title?

Full name	Email Address	LinkedIn Profile
Anthony Alcaraz Author of Agentic Graph RAG (O'Reilly)	alcarazanthony1@gmail.com	https://www.linkedin.com/in/anthony-alcaraz-b80763155/